

Monday - March - 17 - 2025 (Section 6)

"Liquid propellant Rocket Engine" (LPRE)

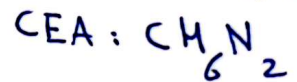
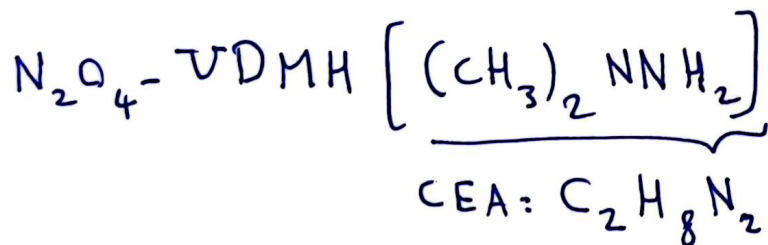
Liquid propellants are divided to monopropellants and bipropellants.

Monopropellants: Capable of igniting or exothermic decomposition (releases heat) sition

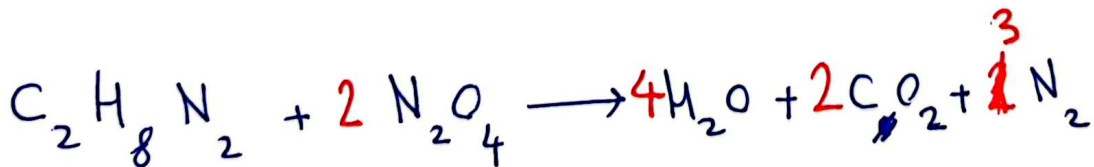
Ex. Hydrazine (N_2H_4): It dissociates by Iodin catalyst, releases heat.

Monopropellants produces small amount of I_{sp} , so they are good for satellites, space probes.

Bipropellants: $O_2(L) - H_2(L)$ / $O_2 - RP1$ / $N_2O_4 - MMH$ [CH_3NHNH_2]



Ex. Consider $N_2O_4 - VDMH$ stoichiometric reaction:



Balance-order: C-H-O-N

$$(P-1) \quad r)_{sto} = OF)_{sto} = \frac{\text{mass of ox}}{\text{mass of fuel}} = \frac{2[28 + 4 \times 16]}{2 \times 12 + 8 \times 1 + 28} = \boxed{3.066}$$

Desirable propellant characteristics:

- ① Low freezing point.
- ② High specific gravity ($SG = \frac{\rho}{\rho_{H_2O}}$)
- ③ stability
- ④ High specific heat ($\bar{c}_p$), high thermal conductivity (k), high boiling point for propellants used for cooling
- ⑤ low vapour pressure is a desirable pumping characteristics: To avoid cavitation (when $p < p_v$, liquid converts to vapour)

(P-2)

Continue of "propellant desired properties"

⑥ Variations of physical properties with temperature should be small and/or similar for fuel and oxidizer.

⑦ readily ignitable.

Recall: Oxidizer-fuel mass ratio ($r \equiv O/F$) or mixture ratio

$$r = \frac{\dot{m}_{ox}}{\dot{m}_f} = \frac{\dot{m}_{ox} \bar{M}_{ox}}{\dot{m}_f \bar{M}_f}$$

Average bulk density (ρ_{bulk})

$$\rho_{bulk} = \frac{m_f + m_{ox}}{V_f + V_{ox}} = \frac{m_f (1 + \frac{m_{ox}}{m_f})}{\frac{m_f}{\rho_f} + \frac{m_{ox}}{\rho_{ox}}} = \frac{(1+r) \rho_f \rho_{ox}}{\rho_{ox} + r \rho_f}$$

Example: Liquid O_2 -Liquid H_2 rocket, $r=4$, $SG_{Lox} = 1.14$

$$SG_{LH_2} = 0.071 \quad \left(SG = \frac{\rho}{\rho_w} = \frac{\rho}{1000} \right) \rightarrow \begin{cases} \rho_{Lox} = 1140 \\ \rho_{LH_2} = 71 \end{cases} \quad \frac{kg}{m^3}$$

$$\rho_{bulk} = \frac{(1+4) 71 \times 1140}{1140 + 4 \times 71} = 284 \quad \frac{kg}{m^3}$$

Example: Lox-RP1 rocket $r=2.58$ $SG_{RP1} = 0.815$

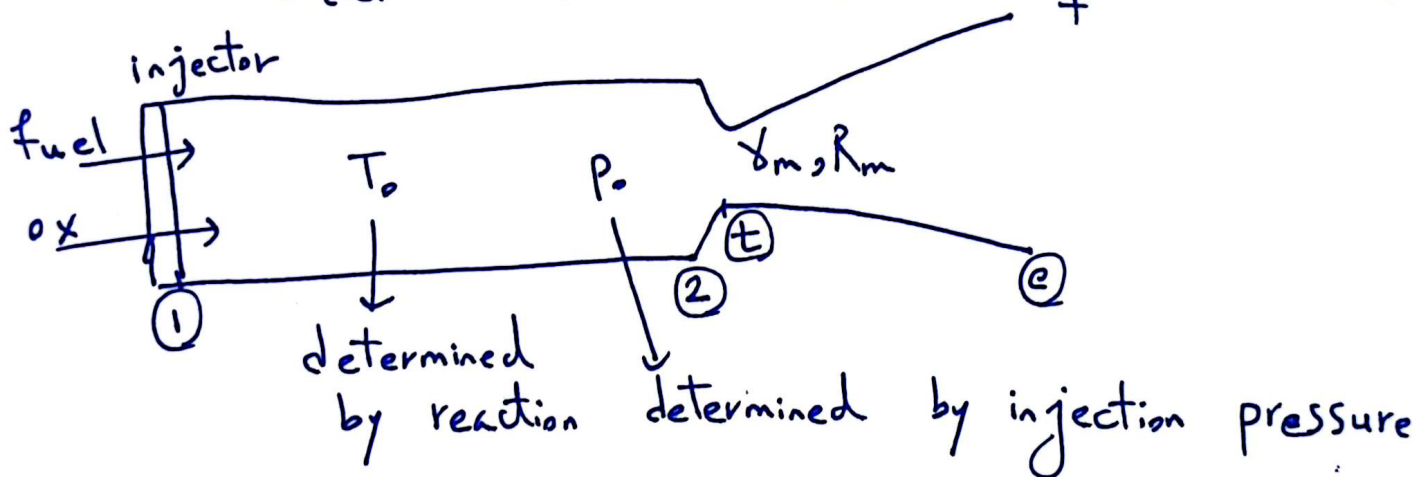
$$\rightarrow \rho_{RP1} = 815 \quad \frac{kg}{m^3}$$

$$\rho_{\text{bulk}} = \frac{(1 + 2.58) 815 (1140)}{1140 + 2.58 \times 815} = 1026 \frac{\text{kg}}{\text{m}^3}$$

For a given propellant mass

$$V_{\text{tank}} \left(\frac{L_{\text{ox}}}{L_{\text{H}_2}} \right) \approx 3.6 V_{\text{tank}} \left(\frac{L_{\text{ox}}}{R_{\text{PI}}} \right)$$

"Consider thrust chamber of liquid rockets"



This thrust chamber needs propellants feed at high ~~temperature~~ pressure.

Unlike solid rockets, there is no increase in p_0 in the combustion chamber.

Recall: $\bar{M}_m = \sum x_i M_i$, $\bar{c}_{p_m} = \sum x_i \bar{c}_{p_i}$ $\gamma_m = \frac{\bar{c}_{p_m}}{\bar{c}_{p_m} - \bar{R}}$

$$R_m = \frac{\bar{R}}{\bar{M}_m}$$

Example: An ideal H_2-O_2 rocket engine has $r = O/F = 5$.
 Propellants are injected at 68 atm and $T = 298.15$ K.
 Use Table 12.2 of textbook to read x_i & T_o of products.

Assuming fully-expanded jet at sea-level ($p_e = p_a = 1$ atm)

Calculate I_{sp} . $\gamma_{mon} = 1.67$, $\gamma_{di} = 1.29$, $\gamma_{tri} = 1.17$

Solution: $p_o = 68 \text{ atm} = 68 \times 0.101325 \text{ MPa} = 6.89 \text{ MPa}$

From table 12.2 of book, $T_o = 3429 \text{ K}$ (CEA, $T_o = 3423 \text{ K}$)

" " " " " $x_{H_2} = 0.3607$
 $x_{O_2} = 0.977 \times 10^{-3}$
 $x_{H_2O} = 0.5744$
 $x_{OH} = 0.0260$
 $x_O = 0.015$
 $x_H = 0.0363$

Read these from CEA as well

$$\bar{M}_m = \sum x_i \bar{M}_i = 0.3607 \times 2 + 0.977 \times 10^{-3} \times 32 + 0.5744 \times 18 + 0.0260 \times 17 + 0.015 \times 16 + 0.0363 \times 1 =$$

$$\boxed{11.59}, \text{ CEA gives } \boxed{\bar{M}_m = 11.685}$$

Combustion chamber

$$R_m = \frac{\bar{R}}{\bar{M}} = \frac{8.3143}{11.59} = 0.7174 \frac{\text{kJ}}{\text{kg K}}$$

(h 2.)

$$\bar{c}_{p_m} = \sum x_i \bar{c}_{p_i}$$

$$\bar{c}_{p_o} = \bar{c}_{p_H} = \frac{\gamma_{mon} \bar{R}}{\gamma_{mon} - 1} = 2.493 \bar{R}$$

$$\bar{c}_{p_{OH}} = \bar{c}_{p_{H_2}} = \bar{c}_{p_{O_2}} = \frac{\gamma_{d_i} \bar{R}}{\gamma_{d_i} - 1} = 4.448 \bar{R}$$

$$\bar{c}_{p_{H_2O}} = \frac{\gamma_{tr_i} \bar{R}}{\gamma_{tr_i} - 1} = 6.882 \bar{R}$$

$$\bar{c}_{p_m} = \sum x_i \bar{c}_{p_i} = 5.772 \bar{R}$$

$$\gamma_m = \frac{\bar{c}_{p_m}}{\bar{c}_{p_m} - \bar{R}} = \frac{5.772}{4.772} = 1.21$$

CEA gives $\gamma_m = 1.15$

$$P_o = 68 \text{ atm}, P_e = 1 \text{ atm}$$

$$\frac{P_o}{P_e} = 68 = \left(1 + \frac{\gamma_m - 1}{2} M_e^2\right)^{\frac{\gamma_m}{\gamma_m - 1}}$$

Solve to get $M_e = 3.21$

CEA gives $M_e = 3.143$

$$T_o = 3429 \text{ K} \rightarrow T_e = \frac{T_o}{1 + \frac{\gamma_m - 1}{2} M_e^2} = 1647 \text{ K}$$

CEA gives $T_e = 1850 \text{ K}$

$$u_e = M_e \sqrt{\gamma_m R_m T_e} = 3.21 \times \sqrt{1.21 \times 717.4 \times 1647} = 3838 \frac{\text{m}}{\text{s}}$$

$$u_{eq} = u_e + \frac{R T_e}{u_e} \left(1 - \frac{P_o}{P_e}\right) = u_e = 3838 \frac{\text{m}}{\text{s}}$$

$$u_{eq})_{CEA} = 3928 \frac{\text{m}}{\text{s}}$$

(P-4)